

Name of the module: Molecular Biology

Number of module: 477-10101

BGU Credits: 6

ECTS credits:

Academic year: 2

Semester: 1

Hours of instruction: 2 hours/week guided-learning study and Zoom discussion, 2 hours/week practical laboratory, 1 hour/week Zoom tutorial

Location of instruction: Guided learning, discussions and tutorials are online. Laboratory sessions in the Medical Faculty Student Labs as scheduled.

Language of instruction: English.

Cycle: B.Med.Sc.

Position: Basic course. A passing mark is obligatory for students in the B.Med.Sc. program.

Field of Education: Molecular Biology subdiscipline of Biochemistry

Responsible department: Shraga Segal Department of Microbiology, Immunology and Genetics

General prerequisites: Biochemistry A+B, Cell Biology

Grading scale: Percentage, with 65% passing mark.

Course Description: A combination of frontal lectures, tutorials, and practical laboratory sessions that teach the fundamentals of molecular biology in theory and practice.

Aims of the module: The purpose of the module is two-fold. 1. To teach the basic cellular machinery involved in information-processing pathways. 2. To teach students fundamental research laboratory practices.

Objectives of the module: each student will 1. Study the basic principles of DNA and RNA transactions in prokaryotic and eukaryotic cells. 2. Study the regulation of gene expression in cells. 3. Be exposed to and understand the use of basic Molecular Biology tools in medical research and medical laboratory diagnostics.

Learning outcomes of the module: On successful completion of the course, the student should be able to:

1. Recount the basic mechanisms and components of information processing in the cell.
2. Discuss how these processes differ between prokaryotes and eukaryotes.
3. Give concrete examples taken from cell-and organism-level processes that illustrate these molecular mechanisms.
4. Be able to properly plan, record and analyze a simple molecular biology research program in the lab.

Attendance regulation: Mandatory for Laboratory sessions and Zoom Discussion sessions. Optional for tutorial sessions.

Teaching arrangement and method of instruction: All teaching except practical laboratory sessions is online. Topics are organised into guided-learning modules that include reading, online teaching resources, assignments, discussion topics and quizzes. Students are organised into teams for group learning and presentation. Interactive Zoom sessions are reserved for team and class discussions.

Lecturer: Dr. Claytus Davis

Contact details:

Davis: Office phone: (08) 6479960

Email: clay@bgu.ac.il

Office hours: Individual Zoom sessions
on an ad hoc basis, Sundays to
Thursday, 9:00 am to 5:00 pm.

Module evaluation: Organized yearly
feedback from students at end of
semester. Periodic review by academic
committee.

Confirmation: the syllabus was
confirmed by the faculty academic
advisory committee to be valid.

Last update: 2020

Assessment:

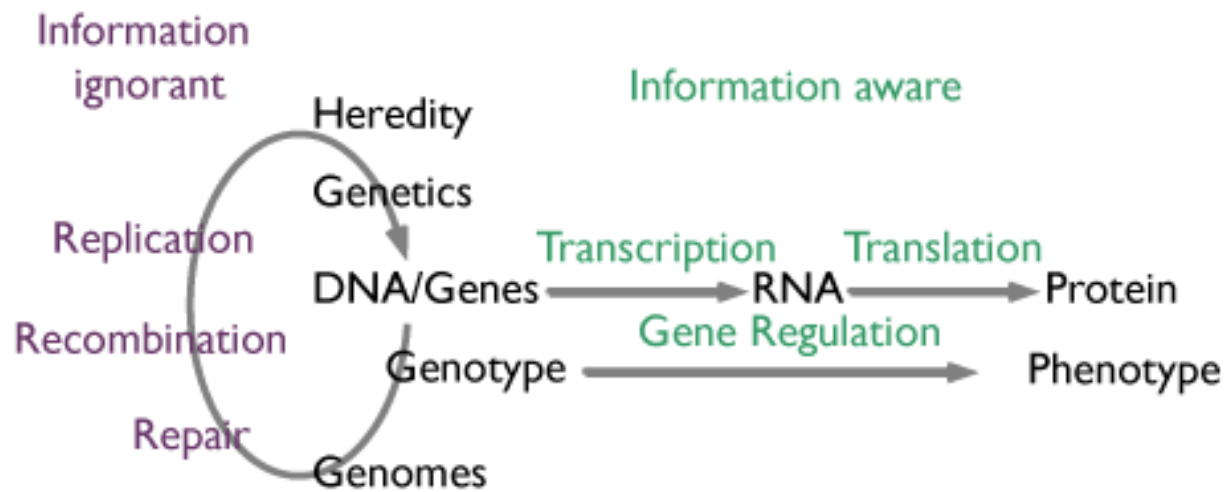
1. Final exam, 40%
2. Laboratory 25% (Exam and Practical, including lab notebook)
3. Quizzes and Assignments 35%

100%

Work and assignments: Students are briefed on the course work, including
quizzes, laboratory notebook, final lab exam and final course exam, at the
beginning of the year.

Time required for individual work: Reading, lab-related work, and material
review is estimated to require 4 - 5 hours/week.

Module Content\ schedule and outlines: The basic molecular processes involved in information processing in the cell are presented to the students in an integrated format in a single diagram (below) referred to throughout the module.



The approximate time for each of the topics, in the order shown, is as follows:

DNA, RNA, chromatin and genomes (6)

Replication (4)

Transcription (4)

Translation (4)

Gene Regulation (8)

Mutation and Repair (4)

Recombination (4)

Cell Cycle (4)

Genetic Engineering (4)

Genome variation (2)

DNA-based Diagnostics (4)

Textbook: **Molecular Biology of the Cell**, ed. **Alberts et al**, 6th edition.

Additional literature and resources: **Posted to course Moodle site and to course web site.**